Algebra 2 — Algebraic fractions · Roots and powers · Indices · Inequalities

No calculator. Marks in brackets. ★ = harder question. Check every solution by substitution.

A · SIMPLIFY AS MUCH AS POSSIBLE

1 $\dfrac{15x^2y}{5x}$ [1]

> $15 \div 5 = 3$, $x^2 \div x = x$: $3xy$.

2 $(8ab^2) \div (12a^2b)$ [1]

> $\dfrac{8ab^2}{12a^2b} = \dfrac{2b}{3a}$ (cancel 4, a and b).

3 $\dfrac{a}{b^2} \times \dfrac{b^3}{c} \times \dfrac{c^2}{a}$ [1]

> Cancel a with a, b^2 with b^3 (leaving b), c with c^2 (leaving c): bc.

4 $\dfrac{4x^2}{y} \div \dfrac{x}{y^2}$ [1]

> Multiply by the reciprocal: $\dfrac{4x^2}{y} \times \dfrac{y^2}{x} = 4xy$.

5 $\dfrac{x}{3} - \dfrac{x}{5}$ [2]

> Common denominator 15: $\dfrac{5x}{15} - \dfrac{3x}{15} = \dfrac{2x}{15}$.

6 $\dfrac{x}{4} + \dfrac{3x}{8}$ [2]

> Common denominator 8: $\dfrac{2x}{8} + \dfrac{3x}{8} = \dfrac{5x}{8}$.

B · EQUATIONS WITH POWERS AND ROOTS

7 Solve $3x^2 - 5 = 70$ [3]

> Add 5: $3x^2 = 75$. Divide by 3: $x^2 = 25$. Square root — *two* answers: $x = 5$ or $x = -5$.
> Check: $3 \times 25 - 5 = 70$ ✓ for both.

8 Solve $\dfrac{x^2 - 6}{2} = 15$ [3]

> Multiply by 2: $x^2 - 6 = 30$. Add 6: $x^2 = 36$, so $x = \pm 6$.

9 Solve $\dfrac{\sqrt{x - 3}}{2} = 2$ [2]

> Multiply by 2: $\sqrt{x - 3} = 4$. Square both sides: $x - 3 = 16$, so $x = 19$. Check: $\sqrt{16} \div 2 = 2$ ✓ (one answer only — $\sqrt{\ }$ means the positive root).

C · INDICES

10 Simplify [3]

a $2(k^4)^3$ **b** $m^7 \div m^2$ **c** $t^3 \times t^9$

> **a** $2k^{12}$ (power of a power: multiply; the 2 is not raised to the power). **b** m^5. **c** t^{12}.

D · INEQUALITIES

11 Solve the inequality $12 \leq 9 - x$ and show the solution on a number line. [3]

> Add x to both sides: $12 + x \leq 9$. Subtract 12: $x \leq -3$. Number line: a **solid** circle at -3 (included) with the arrow pointing left.

12 List the integer solutions of $2 \leq 2x < x + 7$. [3]

> Split into two: $2 \leq 2x \Rightarrow x \geq 1$; and $2x < x + 7 \Rightarrow x < 7$. So $1 \leq x < 7$: the integers are **1, 2, 3, 4, 5, 6** (7 is not included).

13 Solve $3(x - 2) > 5(x - 4)$ [2]

> Expand: $3x - 6 > 5x - 20$. Subtract $3x$: $-6 > 2x - 20$. Add 20: $14 > 2x$, so $x < 7$.

14 Solve $-2x + 5 \geq 11$ [2]

> Subtract 5: $-2x \geq 6$. Divide by -2 — dividing by a negative **reverses** the sign: $x \leq -3$. Check with $x = -4$: $8 + 5 = 13 \geq 11$ ✓; with $x = 0$: $5 \geq 11$ ✗.

Total: 29 marks

Check: cancel only common *factors* · $x^2 = 25$ has two solutions · multiplying or dividing an inequality by a negative flips the sign · solid circle for $\leq$, open circle for $<$.